UNIT II

Data Collection and Management: Introduction, Sources of data, Data Collection and APIs, Exploring and fixing data, Data storage and management, using multiple data sources.

1. Introduction to Data Collection and Management

What is Data Collection?
It is the process of gathering and measuring information on variables of interest in a systematic way to answer questions, test hypotheses, and evaluate outcomes.

What is Data Management?
Data management involves organizing, storing, maintaining, and processing collected data to ensure its quality, integrity, and usability for analysis or decision-making.

Purpose:

- Support decision-making
- Enable analytics and machine learning
- Ensure data is accurate and consistent
- Make data easily accessible and secure

2. Sources of Data
There are several types of sources from where data can be collected:

a. Primary Data Sources
Data collected directly by the user or organization for a specific purpose.
Examples:

- Surveys
- Interviews
- Experiments
- Observations

b. Secondary Data Sources
Data that already exists and is collected by someone else.
Examples:

- Government databases (e.g., Census data)
- Research publications
- Open datasets (e.g., Kaggle, UCI Repository)

- Websites and blogs

c. Web and Social Media

- Twitter, Facebook, Instagram
- News websites
- E-commerce platforms (e.g., Amazon product reviews)

d. Sensors and IoT Devices

- GPS, temperature, traffic sensors
- Smart watches, health trackers

e. Transactional Systems

- Bank systems, POS (Point of Sale) machines
- ERP, CRM software

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3. Data Collection and APIs

a. Manual Data Collection

- Data entry, online forms, spreadsheets
- Time-consuming and error-prone

b. Automated Data Collection

- **Web Scraping**: Extracting data from websites using tools like BeautifulSoup, Scrapy
- **Sensor Data**: Devices automatically send data to servers

c. APIs (Application Programming Interfaces)

APIs allow programs to communicate with each other and fetch real-time data from various services.

Examples:

Source	API Example
Twitter	Twitter API (for tweets)
Weather	Open Weather Map API
Financial	Alpha Vantage, Yahoo Finance API
Maps	Google Maps API

4. Exploring and Fixing Data (Data Cleaning)

Raw data is often incomplete, inconsistent, or inaccurate. Hence, **exploration and cleaning** is essential.

a. Exploring Data

Use tools like:

- **Pandas** (Python) for exploring tables (df.head(), df.describe(), df.info())
- **Visualizations** (matplotlib, seaborn)
- Summary statistics

b. Fixing Issues

Common issues:

Problem	Solution
Missing values	Fill (mean, median, mode) or drop rows/columns
Duplicates	Remove duplicate entries
Incorrect data types	Convert to appropriate formats
Outliers	Identify using z-score or IQR, and handle accordingly
Inconsistent formats	Standardize (e.g., dates, units, text)

6. Using Multiple Data Sources

Often, a single source is not enough, and multiple datasets must be combined.

a. Data Integration

- Combine multiple datasets from different sources
- Join or merge tables (using keys like user ID, timestamp, etc.)

b. Challenges in Combining Multiple Sources

Challenge	Solution
Different formats	Convert all to a standard format
Inconsistent columns/labels	Rename and align structure
Duplicate entries	Remove or consolidate duplicates
Missing values	Impute or fill intelligently